

MILESTONE 3: DAX & BUSINESS METRICS

Project: HotelRevAI – AI Driven Revenue Analysis for Hotels)

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1. Introduction

Milestone 3 focuses on the implementation of DAX (Data Analysis Expressions) and Business Metrics to enhance analytical capabilities of the Hotel Revenue Analysis project. While Milestone 2 established data loading and data modelling foundations, this phase emphasizes the creation of intelligent measures and KPIs that support business decision-making.

In the hotel industry, raw booking data alone is not sufficient to derive insights related to revenue performance, customer behaviour, and pricing optimization. Therefore, advanced DAX measures were created in Power BI to calculate key performance indicators such as Total Revenue, Average Daily Rate (ADR), Revenue per Night, Repeat Guest Ratio, and Occupancy Proxy.

These business metrics transform transactional booking data into meaningful insights that help hotel management understand demand patterns, revenue trends, and customer loyalty behaviour.

2. Data Loading (Recap)

The hotel booking dataset was imported into Power BI using the Excel/CSV data source option. Power Query Editor was used to perform data transformation and cleaning before loading the data into the model. As mentioned in Milestone 2, the dataset contains booking-level records for both City Hotel and Resort Hotel including arrival date, lead time, customer type, ADR, and stay duration.

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The following preprocessing steps were performed:

- Removed null and inconsistent values
- Verified and corrected data types (Date, Numeric, Text)
- Created Total Guests column (Adults + Children + Babies)
- Created Total Nights column (Weekend Nights + Weekday Nights)

- Created Revenue column using ADR and stay duration
- Converted separate date columns into a single Arrival Date column

These steps ensured data accuracy before applying DAX calculations.

3. Data Modelling for Analytical Framework

A star schema data model was implemented to enable efficient filtering and aggregation across dashboards. The model consists of one fact table and multiple dimension tables.

3.1 Fact Table (Fact_Bookings)

The Fact_Bookings table contains transactional booking data such as:

- Arrival Date
- ADR (Average Daily Rate)
- Lead Time
- Customer Type
- Reserved Room Type
- Market Segment
- Distribution Channel
- Total Guests
- Total Nights
- Revenue

This table acts as the central table for all calculations and KPIs.

3.2 Dimension Tables

To support multi-dimensional analysis, the following dimension tables were created:

- Dim_Date (Calendar Table – Year, Month, Quarter)
- Dim_Customer (Customer Type)
- Dim_Country (Geographical Analysis)

- Dim_Hotel (Hotel Type)
- Dim_MarketSegment (Market Segmentation)

A one-to-many relationship was established between:

Dim_Date[Date] → Fact_Bookings[Arrival Date]

This relationship enables accurate time-based analysis across all dashboards.

3.3 Measures Table (DAX Table)

A dedicated Measures Table was created using:

Measures_Table = DATATABLE("Measure", STRING, {"KPI"})

This table is used only to store DAX measures for better model organization and professional dashboard design.

4. DAX Measures Implementation

DAX measures were created to convert raw booking data into meaningful business indicators.

4.1 Revenue Measures

Total Revenue

Total Revenue = SUM(Fact_Bookings[Revenue])

This measure calculates the total income generated from all hotel bookings.

Revenue per Night

Revenue per Night =
DIVIDE([Total Revenue], [Total Nights], 0)

This metric helps evaluate revenue efficiency per occupied night.

4.2 Booking Performance Measures

Total Bookings

Total Bookings = COUNTROWS(Fact_Bookings)

Represents the total number of bookings recorded in the dataset.

Total Guests Count

Total Guests Count = SUM(Fact_Bookings[Total Guests])

Provides insight into total customer volume.

4.3 Customer Behaviour Measures

Repeated Customers

Bookings_Repeated =
CALCULATE(
 COUNTROWS(Fact_Bookings),
 Fact_Bookings[is_repeated_guest] = 1
)

Repeat Guest Percentage

Repeat % =
DIVIDE([Bookings_Repeated], [Total Bookings], 0)

This KPI measures customer loyalty and retention rate.

4.4 Pricing and Demand Metrics

Average Daily Rate (ADR)

Average ADR = AVERAGE(Fact_Bookings[adr])

ADR reflects the average revenue earned per occupied room.

Average Lead Time

Average Lead Time = AVERAGE(Fact_Bookings[lead_time])

This metric helps analyse early booking behaviour.

4.5 Occupancy and Optimization Metrics

Occupancy Proxy

Occupancy Proxy =
DIVIDE([Total Guests Count], [Total Nights], 0)

This is a calculated indicator used as an approximate occupancy measure.

Business metrics were designed to align with real-world hotel management decision-making.

5.1 Revenue Performance Metrics

- Total Revenue – Measures overall financial performance
- Revenue per Night – Evaluates pricing efficiency
- ADR – Indicates pricing strategy effectiveness

These metrics help answer:

- Which period generates maximum revenue?
 - Is the pricing strategy stable over time?
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5.2 Customer Behaviour Metrics

- Total Customers
- Repeated Customers
- Repeat Guest %
- Average Lead Time

These metrics provide insights into:

- Customer loyalty trends
 - Early vs last-minute booking behaviour
 - Demand forecasting patterns
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5.3 Operational Performance Metrics

- Total Bookings
- Total Guests
- Occupancy Proxy

These KPIs help hotel management understand:

- Demand volume
- Room utilization efficiency

- Peak booking periods

6. Integration with Three Dashboard Strategy

The DAX measures were strategically mapped to the three dashboards:

Dashboard 1 – Revenue Performance Overview

Uses:

- Total Revenue
- Total Bookings
- Average ADR
- Total Guests
- Monthly Revenue Trend

This dashboard answers:
“What is happening in hotel revenue performance?”

Dashboard 2 – Customer & Booking Behaviour

Uses:

- Total Customers
- Repeat Guests %
- Average Lead Time
- Bookings by Customer Type

This dashboard explains:
“Why revenue patterns occur based on customer behaviour.”

Dashboard 3 – Revenue Optimization & AI Insights

Uses:

- Revenue per Night
- Occupancy Proxy

- ADR Trend
- Revenue vs Lead Time

7. Conclusion

Milestone 3 successfully enhanced the analytical depth of the project through the implementation of advanced DAX measures and business metrics in Power BI. The creation of structured KPIs enabled effective transformation of raw hotel booking data into actionable business insights.

By integrating revenue metrics, customer behaviour indicators, and optimization measures, the dashboards now support data-driven decision-making for hotel management. The use of DAX improved model efficiency, accuracy of calculations, and dynamic filtering across visualizations.

Overall, Milestone 3 establishes a strong analytical layer that supports the final AI-driven revenue optimization objective of the HotelRevAI project.